SolarEase

Product Requirement & Client Confirmation Document

Purpose:
This document summarizes the current understanding of the SolarEase application based on the discussions and reference websites shared so far. It is intended for **client review and confirmation** before proceeding with further development.

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1. Project Overview

SolarEase is a mobile application intended to help users:

- Track their electricity usage
- Understand their electricity bills
- Estimate electricity costs
- Identify ways to reduce electricity consumption
- Ask questions about electricity and energy
- Analyze electrical appliances
- Explore solar solutions for their home

Overall concept

Understand your electricity → Reduce your consumption → Explore solar

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2. User Login

When the user opens the application, they will see a simple welcome/login screen.

Expected options

- User Name
- **Continue with Google**
- Facebook Login
- Phone OTP

Current status

- **Google Sign-In:** Implemented and working
- **Facebook Sign-In:** To be considered

- **Phone OTP:** Existing implementation will be preserved separately for future use

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3. Home / Dashboard

After successful login, the user will enter the main dashboard.

The dashboard will contain the major SolarEase modules:

Main Modules

1. ⚡ **Bill Calculator**
2. 🤖 **Electricity Assistant / Chatbot**
3. 📷 **Energy Lens**
4. 🧾 **EB Bill Status**
5. 🌞 **Solar Panel**

The dashboard may also display a quick summary of the user's electricity information when official electricity data becomes available.

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4. Bill Calculator

The Bill Calculator will allow users to estimate their electricity bill based on their electricity consumption.

The reference website provided for this project calculates an estimated electricity bill based on **units consumed** and provides a breakdown of the applicable charges. It also explains consumption through meter readings and provides energy-saving information. Reference: TNEB Bill Calculator

Proposed functionality

The user could enter:

Units consumed: 400 units

The application can display:

- Estimated bill amount
- Units consumed
- Slab-wise calculation
- Applicable charges

- Estimated total
- Energy-saving suggestions

Possible future functionality

If official EB data becomes available, the calculator could use the user's actual consumption instead of requiring manual entry.

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5. Electricity Assistant / Chatbot

SolarEase will provide an AI-based electricity assistant.

Users can ask questions related to:

Bills

"What was my last month's bill?"

"Why is my bill higher this month?"

Consumption

"How many units did I consume?"

"How much electricity am I using?"

Energy Saving

"How can I reduce my electricity bill?"

"Which appliances consume more electricity?"

"How can I save electricity at home?"

Solar

"How can I install solar at my house?"

"How much solar do I need?"

"Will solar reduce my electricity bill?"

Personalized assistance

If official electricity data is available, the chatbot should ideally be able to answer questions using the user's own electricity information.

For example:

User: Why is my bill higher this month?

The chatbot could compare available consumption/billing information and provide an explanation.

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6. Energy Lens

The Energy Lens will be a camera-based feature designed specifically for identifying **electrical appliances**.

The user can point the camera toward a room or area of their house.

Example

The Lens could identify:

Fan
Approx. 75 W

Television
Approx. 100 W

LED Bulb
Approx. 10 W

The purpose is to help the user understand the approximate electrical load of the appliances around them.

Possible additional functionality

The Lens may eventually estimate:

- Approximate power consumption
- Daily consumption
- Monthly consumption
- Approximate electricity cost
- Appliances consuming more electricity
- Energy-saving recommendations

Note: Appliance wattage detected through the camera would be an estimate unless the exact appliance model/specification can be identified.

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7. EB Bill Status

SolarEase is intended to allow users to view their electricity/bill information within the application.

The official TNPDCL Bill Status page currently requires information such as:

- Service Connection Number
- Registered Mobile Number
- Captcha

to access bill information.
Reference: TNPDCL Bill Status

Possible information in SolarEase

Depending on the officially available data, SolarEase could display:

- Current bill
- Previous bill
- Units consumed
- Previous consumption
- Meter readings
- Due date
- Payment status
- Billing history
- Consumption history

Important dependency

The client has indicated that SolarEase may request official access to electricity data as a startup/project.

The exact method of obtaining this data is **not yet finalized**.

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8. Solar Panel

The Solar Panel module is intended to come **after the user understands their electricity consumption**.

The idea is:

Understand electricity usage first → Then decide whether solar is useful.

Possible functionality

The Solar module could provide:

- Recommended solar capacity
- Estimated electricity generation
- Estimated savings
- Approximate installation cost
- Estimated payback period
- Solar information
- Solar benefits
- Government subsidy information, if applicable

Possible user journey

Electricity Consumption

↓

Understand Usage

↓

Calculate / Analyze Bill

↓

Find Ways to Save

↓

Explore Solar

↓

Solar Recommendation

Whether SolarEase will also connect users with solar installation providers is still to be confirmed.

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9. Government Electricity Data Integration

This is an important part of the SolarEase concept.

The application is expected to eventually allow users to access their electricity information through an authorized connection with the relevant electricity/government system.

Possible user data

Depending on the approved integration, this may include:

- Consumer information
- Service Connection Number
- Meter information
- Current consumption
- Previous consumption
- Current bill
- Previous bills
- Due dates
- Payment/bill status
- Consumption history

Expected user linking process

One possible flow could be:

Login

↓

Enter Service Connection Number

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Verify Registered Mobile Number

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Link EB Account

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Fetch Electricity Data

This is only a proposed flow and should not be considered finalized until the client confirms the official integration process.

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10. User Data & Privacy

Since SolarEase may handle personal electricity and billing information, user data must be handled carefully.

The application may need to store:

- User profile
- Mobile number
- Email
- Electricity account information
- Electricity consumption history
- Bill history

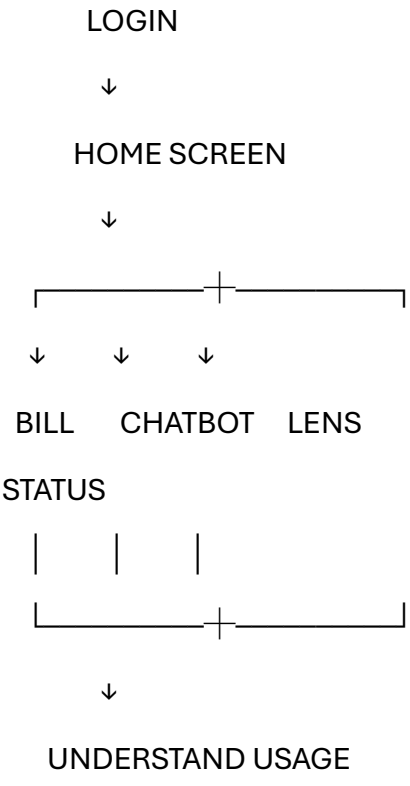
The user's registered mobile number is expected to be important for identifying/verifying the user's electricity account and protecting their data.

The exact data storage and access rules will depend on the permissions and requirements of the electricity authority.

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11. Overall SolarEase User Journey

The proposed overall experience is:



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FIND WAYS TO SAVE

↓

EXPLORE SOLAR

↓

SOLAR RECOMMENDATION

The main objective is to help users move from:

"What am I consuming?"

to

"Why am I consuming this much?"

to

"How can I reduce it?"

and finally:

"Should I consider solar?"

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12. Current Feature Status

Feature	Current Status
Google Sign-In	✅ Working
Facebook Sign-In	🟡 To be confirmed
Phone OTP	🟢 Preserved for future use
Home Dashboard	🔄 In development
Bill Calculator	🔄 Requirement confirmation
Electricity Chatbot	🔄 Requirement confirmation
Energy Lens	🔄 Requirement confirmation
EB Bill Status	🔄 Depends on official data access
Solar Panel	🔄 Requirement confirmation

Feature	Current Status
Government EB Integration	🟡 Requires clarification

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13. Client Questions & Confirmation

This section is intended for the client to review and correct.

A. Login

Q1. Should Google Sign-In be available?

Our understanding: Yes.
Status: 🟢 Confirmed

Q2. Should Facebook Sign-In be available?

Our understanding: It may be required, but this needs confirmation.
Status: 🟡 Need confirmation

Q3. Should Phone OTP remain in the project?

Our understanding: Yes. The existing OTP implementation should be preserved separately so it can be enabled/used when required.

Status: 🟢 Confirmed

B. Government Electricity Data

Q4. Will SolarEase receive official access to electricity/EB data?

Our understanding: Mostly yes. The client mentioned that access may be requested as part of the startup/project.

Status: 🟡 Need final confirmation

Q5. Will the government/electricity authority provide an API?

Our understanding: Unknown.

Status: ❓ Need confirmation

Q6. What information will the government integration provide?

For example:

- Current bill
- Previous bills
- Units consumed
- Meter readings
- Due date
- Payment status
- Consumption history

Our understanding: Not yet known.

Status: ❓ Need confirmation

Q7. Will the user need to provide their Service Connection Number?

Our understanding: Most likely yes.

Status: 🟡 Need confirmation

Q8. Will the user's registered mobile number be used for verification?

Our understanding: Yes. The user's mobile number is important for identifying/verifying the electricity account and protecting user data.

Status: 🟢 Confirmed / expected

Q9. Can SolarEase store the user's electricity and billing history?

Our understanding: Yes. This would allow the application to provide personalized information and chatbot responses.

Status: 🟡 Subject to government/data-access permissions

Q10. How will the user link their EB account to SolarEase?

Our proposed understanding:

Login

↓

Enter Service Connection Number

↓

Verify Registered Mobile Number

↓

Link EB Account

↓

Fetch EB Data

Status: ❓ Need confirmation

C. Bill Calculator

Q11. Should users be able to manually enter units and calculate an estimated bill?

Our understanding: Yes.

Status: 🟡 Need confirmation

Q12. Should the calculator automatically use the user's actual EB consumption when government data becomes available?

Our understanding: This would be useful and fits the overall concept.

Status: ❓ Need confirmation

D. Electricity Chatbot

Q13. Should the chatbot answer general electricity-related questions?

Our understanding: Yes.

Status: 🟡 Need confirmation

Q14. Should the chatbot use the user's personal electricity information?

For example:

"What was my last month's bill?"

"How many units did I consume?"

"Why did my bill increase?"

Our understanding: Yes. This appears to be one of the main purposes of the chatbot.

Status: 🟡 Need confirmation

Q15. Should the chatbot provide electricity-saving recommendations?

Our understanding: Yes.

Status: 🟡 Need confirmation

Q16. Should the chatbot answer solar-related questions?

Our understanding: Yes.

Status: 🟡 Need confirmation

E. Energy Lens

Q17. Should the Lens identify electrical appliances using the camera?

Our understanding: Yes.

Status: 🟡 Need confirmation

Q18. Should it display the approximate wattage of detected appliances?

Example:

Fan — ~75 W
TV — ~100 W

Our understanding: Yes.

Status: 🟡 Need confirmation

Q19. Should it calculate approximate energy consumption from detected appliances?

Our understanding: This appears to be intended, but needs confirmation.

Status: ❓ Need confirmation

Q20. Should it calculate approximate monthly electricity cost?

Our understanding: This could be a useful extension.

Status: ❓ Need confirmation

F. Solar Panel

Q21. Should SolarEase recommend solar based on the user's electricity consumption?

Our understanding: Yes. The user should understand their electricity consumption first and then explore solar if interested.

Status: 🟡 Need confirmation

Q22. Should the Solar module calculate the required solar capacity?

Our understanding: Likely yes.

Status: ❓ Need confirmation

Q23. Should it estimate the user's potential savings from solar?

Our understanding: Likely yes.

Status: ❓ Need confirmation

Q24. Should it estimate installation cost and payback period?

Our understanding: This could be included, but it has not yet been confirmed.

Status: ❓ Need confirmation

Q25. Should SolarEase connect users with solar installation providers?

Our understanding: Unknown.

Status: ❓ Need confirmation

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14. Final Confirmation

Once the above questions are answered, the confirmed requirements can be used as the **official scope for the next stage of SolarEase development**.

Most important points to clarify first

1. **How SolarEase will officially receive EB data**
2. **Whether an API will be provided**
3. **What data the API will provide**
4. **How users will link their EB account**
5. **Exactly what the chatbot should be able to access**
6. **How advanced the Energy Lens should be**
7. **Exactly what the Solar module should calculate/provide**
8. **Whether Facebook Login is required**